INCH-POUNDS

MIL-C-71042 (AR)
14 Feburary 1992

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, LOGIC: 11784523

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the Logic Circuit Card Assembly which is a component in the Electronic Control Unit (ECU). The ECU is part of the Target Holding Mechanism/Tank Gunnery (THM/TG). The Logic Circuit Card Assembly provides the routing of correct signals within the ECU which controls the various operations of the THM/TG (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (See 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, U.S Army ARDEC, ATTN SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A

FSC 5999

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

11784523 - Circuit Card Assembly, Logic
11784525 - Schematic Diagram, Logic
11785556 - Soldering Requirements for Electrical and Electronic Assemblies (RETS)

SPECIAL PACKAGING INSTRUCTIONS

PS11784523 - Packaging of Logic Circuit Card

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 11784523, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the Logic Circuit Card Assembly shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

3.3 Environmental service conditions.

3.3.1 Vibration. The Logic Circuit Card Assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.6 after exposure to sinusoidal vibrations for 1 hour along the vertical axis within the frequency range of 10 to 40 hertz (Hz). The rate of change of frequency at each frequency range shall not exceed 1 Hz per second.

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 40 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 60 minutes.
The sweep rate shall be 3.75 minutes ± 10%.
The 10-40-10 hertz sweep time is 2 minutes 30 seconds ± 10%.

3.3.2 Temperature.

3.3.2.1 Operating. The Logic Circuit Card Assembly shall meet the requirements of 3.4 and 3.6 when stabilized thermally at any temperature between -25 degrees F and + 125 degrees F.

3.3.2.2 Non-operating (storage). The Logic Circuit Card Assembly shall not suffer permanent damage when subjected to the constant temperature extremes of -40 degrees F for a period of 4 hours and +160 degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.6 when returned to room ambient.

3.4 Electrical. The electrical characteristics and continuity of the Logic Circuit Card Assembly shall be as specified on the Logic Schematic Diagram drawing 11784525. Refer to this schematic diagram for the performance requirements of 3.6 and performance testing of 4.7.3.

3.5 Circuit identification. The contact fingers (Pins) of the Circuit Board shall have adequate permanent labeling for proper identification of Pins in the circuitry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

3.6 Performance.

3.6.1 Functional description. The Logic Circuit Card Assembly provides the routing of signals received by the Electronic Control Unit (ECU) of the Target Holding Mechanism/Tank Gunnery (THM/TG) to control the desired position of the Target. The Logic Circuit Card Assembly provides the necessary voltage to the Target Lifting Mechanism, and introduces delays which allow target to raise and lower in accordance with the function switch setting of the ECU.

3.6.2 Supply voltage. The Logic Circuit Card Assembly shall operate when +12 Volt Direct Current (Vdc) is connected to contact Pin Z, and power supply return Ground (GND) to contact Pin A.

3.6.3 Raise target mode. With + 12 Vdc applied at contact pin E the following voltages shall be present:

Pin N	+1.5 Vdc Max
Pin M	+0.75 Vdc Max

With Pin P shorted to GND and contact Pin C open the following voltages shall be present:

Pin N	+12 ± 0.1 Vdc
Pin M	+12 ± 0.1 Vdc

3.6.4 Lower target mode. With + 12 Vdc applied at contact pin K the following voltages shall be present:

Pin N	+1.5 Vdc Max
Pin R	+1.5 Vdc Max

With Pin C shorted to GND and contact Pin P open the following voltages shall be present:

Pin N	+12 ± 0.1 Vdc
Pin R	+12 ± 0.1 Vdc

3.6.5 Lifting mechanism failure. With ± 12 Vdc applied at contact pin E, the following voltages shall be present after 11 ± 2 seconds:

Pin N	+1.5 Vdc Max
Pin M	+0.75 Vdc Max

When +12 Vdc is applied at contact Pin K, and after 11 ± 2 seconds the following voltages shall be present:

Pin N	+12 ± 0.1 Vdc
Pin R	+12 ± 0.1 Vdc

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

3.6.6 Bad hit sensor test. This requirement simulates short and open crystal conditions. With the circuit card in the condition of 3.5.2 and 3.5.3, and contact Pin V shorted to GND, the following voltages shall be present:

Pin N	+12 ± 0.1 Vdc
Pin R	+12 ± 0.1 Vdc

3.6.7 Raising target mode with simulated bad crystal. With the circuit card in the initial conditions of 3.5.2 and 3.5.3. The following voltages shall be present:

Pin N	+12 ± 0.1 Vdc
Pin M	+12 ± 0.1 Vdc

3.6.8 Function switch test. With +12 Vdc applied to contact Pin Z, contact Pin V shall be open (floating), contact Pin P shall be shorted to GND, and contact Pin C shall be open.

3.6.8.1 Function switch in position 1 (simulated). With the circuit card set as in 3.6.8, perform the following steps and obtain the required results:

a. When +12 Vdc is applied at contact Pin E, Voltage at Pin B shall be +1.0 mVdc Max
b. When a pulse of +12 Vdc amplitude and 1.0 ms pulse width is applied at contact Pin X, the following voltages shall be present:

Pin B	+11.6 Vdc Min
Pin N	+12 ± 0.1 Vdc
Pin M	+12 ± 0.1 Vdc

c. When a pulse of +12 Vdc amplitude and 1.0 ms pulse width is applied at contact pin X, the pulse at Pin Y shall be +12 Vdc amplitude with a duration of 39 ±5 ms.
d. When a pulse of +12 Vdc amplitude and 1.0 ms pulse width is applied at contact pin X, the pulse at Pin U shall be +12 Vdc amplitude with a duration of 3.5 ± .5 ms.

3.6.8.2 Function switch in position 2 (simulated). With the circuit card set as in 3.6.8, perform the following steps and obtain the required results:

a. Short contact Pin S to GND, apply +12 Vdc at contact Pin E. When a pulse of +12 Vdc amplitude and 1.0 ms pulse width is applied at contact Pin X, the following voltages shall be present:

Pin B	+11.6 Vdc Min after 11 ± 2 Secs
Pin N	+12 ± 0.1 Vdc
Pin R	+12 ± 0.1 Vdc

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

b. Short contact Pin J to GND. When +12 Vdc pulse is applied at pin X, the following voltages shall be present:

Pin B	+1.0 mVdc Max
Pin N	+12 ± 0.1 Vdc
Pin R	+12 ± 0.1 Vdc
Pin M	+12 ± 0.1 Vdc

3.6.8.3 Function switch in position 3 (simulated). With the circuit card set as in 3.6.8, contact pin S open and contact Pin L shorted to GND, apply the +12 Vdc pulse at Pin X. Observe and record the sequence of events at Pins R, N, M, B, X, C, and P, which shall be in accordance with Figure 1.

3.7 Soldering. The circuit board shall be soldered per the requirements of 11785556.

3.8 First article. When specified in the contract or purchase order, sample shall be subjected to first article inspection in accordance with the technical provisions herein (See 4.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

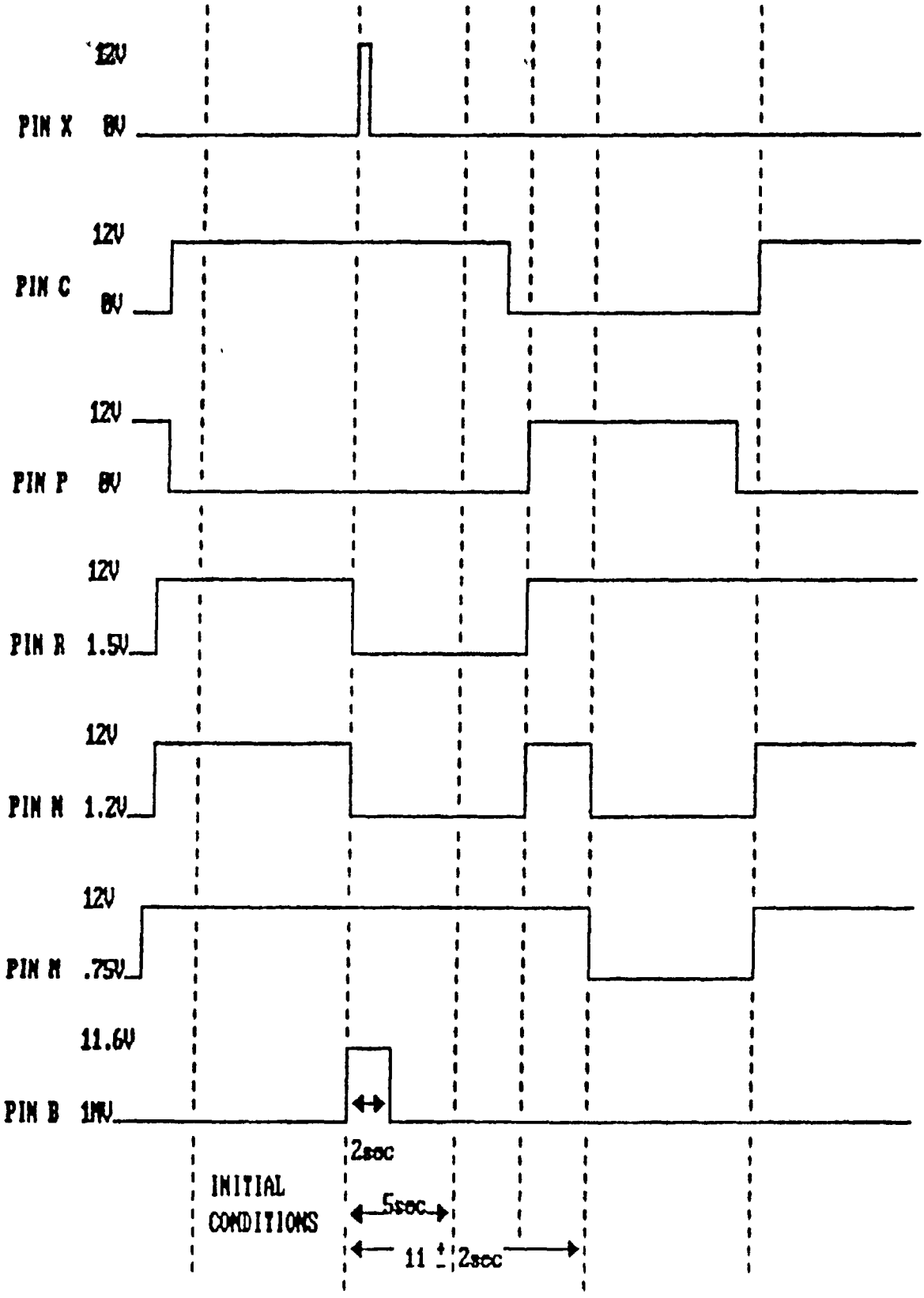


Figure 1. Voltage Outputs per Paragraph 3.6.8.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+60 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) Logic Circuit Card Assemblies for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table I (Functional test) and Table II (Special Sampling) herein and to any or all requirements of the applicable drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 shall be performed as specified in 4.4.2.3 and 4.5.

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis and 4.5.

TABLE I. Functional tests
(100% Inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.6	4.7.3
103. Soldering	3.7	4.8
104. Workmanship	3.2d	Visual
Minor:		
201. Identification	3.5	4.9

4.5 Special sampling. One Logic Circuit Card Assembly out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table I and shall then meet the tests in Table II. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.

TABLE II. Special sampling

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
105. Vibration	3.3.1	4.7.1.1
106. Temperature	3.3.2	4.7.1.2
Minor:	None	N/A

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the components inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table III (Test equipment list), or equivalent meeting minimum requirements, shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

TABLE III. Test equipment list

ITEM NUMBER	DESCRIPTION
1. DC Power Supply	+12 Vdc ± 0.1 Vdc. HP Model 6284A or equivalent
2. Oscilloscope	Dual time base with mixed sweep capabilities. Input impedance of 1 megohm or greater, 0 - 15 Vdc range. Tektronix 2235 or equivalent
3. Generator	Produces a 100 kHz ± 5 % square wave with an amplitude of 15 Vdc into a 25 ohm load. Wavetek Model 145 or equivalent
4. Digital Multimeter	Accuracy and resolution of ± 0.2 % from 1 - 100 Vdc. Input Impedance 10 megohms or greater (2 required). Fluke Model 8060A or equivalent
5. Resistors	180 ohms ± 2%, 2 Watt (4 required)
6. Temperature Chamber	-40 deg C to +71 deg C at a rate of ± 6 deg C/minute minimum

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The Logic Circuit Card Assembly shall be secured to the vibration generator using a suitable fixture in a vertical position with the connector fingers pointing to the right when viewing the component side of the board. After vibration, the Logic Circuit Card Assembly shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1. If the requirements are not met, the item shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

4.7.1.2 Temperature.

4.7.1.2.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.2. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.2. If the requirements are not met, the item shall be rejected.

4.7.1.2.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.2. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.2. If the requirements are not met, the item shall be rejected.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4. If the requirements are not met, the item shall be rejected.

4.7.3 Performance. The performance of the Logic Circuit Card Assembly shall be tested using the equipment listed in Table III. All input and output parameters shall comply with the requirements of 3.6. If the requirements are not met, the item shall be rejected.

4.7.3.1 Step-by-step procedure. The following test procedure shall be conducted in order to comply with the requirements of 3.6. If the requirements are not met, the item shall be rejected.

4.7.3.1.1 Initial conditions.

a. Install a 180 ohm resistor each between pins R, N, and M of Logic circuit card and + 12 Vdc.

b. Install a 180 ohm resistor between pin B and ground.

c. Apply + 12 Vdc $\pm$ 0.1 Vdc to pin Z.

d. Connect Power Supply return (ground) GND to pin A.

e. Pins P, V, S, L, K, D, X, and J are open (floating).

f. Install a jumper wire between pin H and GND.

g. Install a jumper wire between pin C and GND.

h. A voltage check at pin F should be + 8.36 $\pm$ 0.1 Vdc.

Notice: All voltages are measured with respect to ground (GND) or power supply return.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

4.7.3.1.2 Raise target. Test to meet the requirements of 3.6.3

a. Momentarily apply + 12 Vdc at pin E and immediately (within 9 seconds max):

b. Measure the voltage at pin N. The voltage shall be + 1.5 Vdc maximum.

c. Measure the voltage at pin M. The voltage shall be + 0.75 Vdc maximum.

d. Short pin P to GND.

e. Open pin C (The target is now raised and stopped in place).

f. Measure the voltage at pin N. The voltage shall be + 12 ± 0.1 Vdc.

g. Measure the voltage at pin M. The voltage shall be + 12 ± 0.1 Vdc.

4.7.3.1.3 Lower target. Test to meet the requirements of 3.6.4.

a. REMOVE + 12 Vdc from pin E.

b. Apply + 12 Vdc to pin K.

c. Measure the voltage at pin N. The voltage shall be + 1.5 Vdc maximum.

d. Measure the voltage at pin R. The voltage shall be + 1.5 Vdc maximum.

e. Short pin C to GND.

f. Open pin P.

g. Measure the voltage at pin N. The voltage shall be + 12 ± 0.1 Vdc.

h. Measure the voltage at pin R. The voltage shall be + 12 ± 0.1 Vdc.

4.7.3.1.4 Lifting mechanism failure. Test to meet the requirements of 3.6.5.

a. Repeat 4.7.3.1.1 c through g.

b. Momentarily apply + 12 Vdc at pin E. Within 11 ± 2 seconds:

c. Measure the voltage at pin N. The voltage shall be + 1.5 Vdc maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

d. Measure the voltage at pin M. The voltage shall be + 0.75 Vdc maximum after 11 ± 2 seconds.

e. Measure the voltage at Pin N. The voltage shall be + 12 Vdc ± 0.1 Vdc.

f. Measure the voltage at Pin M. The voltage shall be + 12 Vdc ± 0.1 Vdc.

g. Remove + 12 Vdc from pin E.

h. Momentarily apply + 12 Vdc at pin K. Within 11 ± 2 seconds:

i. Measure the voltage at pin N. The voltage shall be + 12 ± 0.1 Vdc.

j. Measure the voltage at pin R. The voltage shall be + 12 ± 0.1 Vdc.

4.7.3.1.5 Bad hit sensor test. (short and open crystal conditions). Test to meet the requirements of 3.6.6.

a. Repeat steps 4.7.3.1.1 c, d, e, h, 4.7.3.1.2 d and e (Raise the target).

b. Short pin V to GND.

c. Measure the voltage at pin N. The voltage shall be + 1.5 ± 0.1 Vdc.

d. Measure the voltage at pin R. The voltage shall be + 1.5 ± 0.1 Vdc.

4.7.3.1.6 Raising target with simulated bad crystal. Test to meet the requirements of 3.6.7.

a. Turn off the power supply.

b. Disconnect pin C from GND.

c. Short pin V to GND.

d. Repeat steps 4.7.3.1.1 c, d, e, h, 4.7.3.1.2 d and e. (raise the target).

e. Measure the voltage at pin N. The voltage shall be + 12 ± 0.1 Vdc.

f. Measure the voltage at pin M. The voltage shall be + 12 ± 0.1 Vdc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

4.7.3.1.7 Function switch test.

4.7.3.1.7.1 Initial conditions.

a. Apply power to the Logic Card.

b. Open pin V. (floating)

c. Short Pin P to GND.

d. Open pin C.

4.7.3.1.7.2 Function switch in position 1 (simulated). Test to meet the requirements of 3.6.8.1.

a. Apply + 12 Vdc at pin E.

b. Measure the voltage at pin B. The voltage shall be + 1.0 mVdc maximum.

c. Apply a pulse of + 12 Vdc amplitude and 1.0 ms pulse width at pin X.

d. Measure the voltage at pin B. The voltage shall be + 11.6 Vdc minimum.

e. Measure the voltage at pin N. The voltage shall be + 12 ± 0.1 Vdc.

f. Measure the voltage at pin M. The voltage shall be + 12 ± 0.1 Vdc.

g. Repeat c.

h. Measure (observe) the pulse at pin Y. The pulse shall be + 12 Vdc amplitude with a duration of 39 ± 5 ms.

i. Repeat c.

j. Measure (observe) the pulse at pin U. The pulse shall be + 12 Vdc amplitude with a duration of 3.5 ± .5 ms.

4.7.3.1.7.3 Function switch in position 2 (simulated). Test to meet the requirements of 3.6.8.2.

a. Repeat 4.7.3.1.7.1 b, c, and d.

b. Short pin S to GND.

c. Apply + 12 Vdc at pin E.

d. Apply a pulse of + 12 Vdc amplitude and 1.0 ms pulse width at pin X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

e. Measure the voltage at pin B. The voltage shall be + 11.6 Vdc minimum. After 11 ± 2 seconds:

f. Measure the voltage at pin N. The voltage shall be + 12 Vdc ± 0.1 Vdc.

g. Measure the voltage at pin R. The voltage shall be + 12 ± 0.1 Vdc.

h. Short pin J to GND.

i. Repeat a, b, c, and d.

j. Measure the voltage at pin B. The voltage shall be + 1.0 mVdc maximum.

k. Measure the voltage at pin N. The voltage shall be + 12 Vdc ± 0.1 Vdc.

l. Measure the voltage at pin R. The voltage shall be + 12 Vdc ± 0.1 Vdc.

m. Measure the voltage at pin M. The voltage shall be + 12 Vdc ± 0.1 Vdc.

4.7.3.1.7.4 Function switch in position 3 (simulated). Test to meet the requirements of 3.6.8.3.

a. Repeat 4.7.3.1.7.1 b, c, and d.

b. Open pin S to GND.

c. Short pin L to GND.

d. Apply a pulse of + 12 Vdc amplitude and 1.0 ms pulse width at pin X. The sequence of events at output pins R, N, M, B, X, C and P shall be in accordance with Figure 1. (3.6.8.3)

4.8 Soldering. Soldering processes shall be inspected to the requirements and acceptance criteria of 11785556 as specified by 3.7 herein.

4.9 Identification. Visual inspect Circuit Boards for adequate labeling of contact pins, as required by 3.5 herein.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS 11784523. The level of protection shall be as specified in the procurement document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Logic Circuit Card Assembly is one of Three (3) Circuit Cards used in the Electronic Control Unit (ECU) of the Target Holding Mechanism/Tank Gunnery (THM/TG).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Applicable stock number.
d. First article sample requirements.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certification of conformance for each lot of shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

a. Electronic Control Unit (ECU)
b. Target Holding Mechanism/Tank Gunnery (THM/TG)
c. Hit Sensor

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 5999-A121)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. **The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.**

   The submitter of this form must complete blocks 4, 5, 6, and 7.

3. **The preparing activity must provide a reply within 30 days from receipt of the form.**

NOTE. This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-C-71042 (AR)	2 DOCUMENT DATE (YYMMDD) 14 FEBRUARY 1992

3 DOCUMENT TITLE
CIRCUIT CARD ASSEMBLY, LOGIC 11784523

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle Initial)	b ORGANIZATION	
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7 DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME US ARMY ARDEC STANDARDIZATION OFFICE	b TELEPHONE (Include Area Code) (1) Commercial: (201) 724-6674 (2) AUTOVON: 880-6674
DDRESS (Include Zip Code) ATTN SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete* 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:19Z
Check the source to verify that this is the current version before use.